Severe Combined Immunodeficiency Due to DNA-PKcs Deficiency — Comprehensive Disease Report

Disease: Severe Combined Immunodeficiency Due To DNA-PKcs Deficiency **Causal gene:** *PRKDC* (Protein Kinase, DNA-Activated, Catalytic Polypeptide), encoding the **DNA-dependent protein kinase catalytic subunit (DNA-PKcs) MONDO:** MONDO:0014423 · **OMIM:** 615966 (Immunodeficiency 26, IMD26) · **Inheritance:** Autosomal recessive

Summary

Severe combined immunodeficiency (SCID) due to DNA-PKcs deficiency is an **ultra-rare, autosomal-recessive inborn error of immunity caused by biallelic loss-of-function or hypomorphic mutations in *PRKDC***, the gene encoding the DNA-dependent protein kinase catalytic subunit (DNA-PKcs). DNA-PKcs is a core factor of the non-homologous end-joining (NHEJ) DNA double-strand-break repair pathway; it recruits and activates the nuclease **Artemis** to open DNA hairpins during V(D)J recombination, the process that builds the antigen-receptor repertoire of T and B lymphocytes. When DNA-PKcs function is lost, V(D)J recombination fails, T- and B-cell development arrests, and patients develop a **radiosensitive T-B-NK+ SCID**. As of 2024, only about **seven patients** have been described worldwide, making this one of the rarest defined SCID genotypes.

The human disease spectrum is broader than classic neonatal SCID. The predominant reported presentation is an **"atypical" / "leaky" form dominated by granulomas, autoimmunity, and inflammation** (5 of 7 patients), featuring cutaneous and visceral granulomas, autoimmune cytopenias and thyroiditis, a strong blood interferon signature, and—in at least one case—persistent **rubella vaccine-strain virus** within granuloma tissue. Mechanistically, this is linked both to failed V(D)J recombination and to a distinct role of DNA-PK in promoting **AIRE-dependent central T-cell tolerance**. Uniquely among NHEJ disorders, DNA-PKcs deficiency can also cause **severe human-specific neurodevelopmental disease** (microcephaly, seizures,

progressive cerebral and hippocampal atrophy) that is not recapitulated in Prkdc-deficient mice, dogs, or horses, revealing a stringent requirement for DNA-PKcs in human neuronal development.

Untreated SCID is usually fatal in the first year of life. **Allogeneic hematopoietic stem cell transplantation (HSCT) is the only established curative therapy**, but patients' intrinsic radio- and chemosensitivity demands minimizing alkylator-based conditioning and ionizing radiation. Survival is best when the diagnosis is made presymptomatically through newborn screening (low/absent T-cell receptor excision circles, TRECs). Diagnosis cannot rely on DNA-PK kinase activity—the recurrent p.L3062R variant retains kinase activity—so **molecular sequencing of *PRKDC* is required**, supported by the characteristic long palindromic (P)-nucleotide signature in antigen-receptor coding joints and by cellular radiosensitivity assays.

Key Findings

F001 — A recurrent hypomorphic missense mutation, p.L3062R, dominates the human cohort

Of the seven reported patients with human DNA-PKcs deficiency, **six carry the autosomal-recessive *PRKDC* missense variant p.L3062R**. The first human case was reported in 2009. The variant is **hypomorphic rather than null**: DNA-PKcs protein is still expressed and retains autophosphorylation and DNA end-binding capacity, but the mutant protein **fails to properly activate the Artemis nuclease**, impairing hairpin opening during V(D)J recombination. Structural studies place residue L3062 in the FAT/kinase region of DNA-PKcs and explain how the substitution produces insufficient Artemis activation while preserving the kinase's own catalytic output.

> *"6 patients displayed the autosomal recessive p.L3062R mutation in PRKDC-encoding DNA-PKcs"* — PMID: 38977084
>
> *"The mutation concerned a hypomorphic missense mutation (L3062R) that did not result in absence of protein expression nor in deficient in vivo or in vitro (auto)phosphorylation"* — PMID: 19823081

> *"All of the findings are useful in explaining how a hypomorphic L3062R missense mutation of DNA-PKcs could lead to insufficient Artemis activation, hence RS-SCID"* — PMID: 35801871

This has a direct **diagnostic corollary**: because the recurrent allele leaves kinase activity intact, a normal DNA-PK kinase assay does not exclude a *PRKDC* defect.

F002 — The predominant human presentation is atypical SCID with granuloma, autoimmunity, and inflammation

In the cohort of seven patients, **atypical (leaky) SCID with inflammatory lesions, granulomas, and autoimmunity was the predominant manifestation (5 of 7)**. Reported features include cutaneous and visceral granulomas, autoimmune cytopenias and thyroiditis, a strong whole-blood interferon signature, progressively declining T- and B-cell counts over time, and severe infections. In two patients tested for rubella, **rubella vaccine-strain virus was detected within granuloma tissue in one**, linking persistent live-attenuated vaccine virus to the granulomatous pathology—a phenomenon well documented across combined immunodeficiencies.

> *"Atypical severe combined immunodeficiency with inflammatory lesions, granulomas, and autoimmunity was the predominant clinical manifestation (n = 5 of 7)"* — PMID: 38977084
>
> *"Rubella viral strain was detected in the granuloma of 1 patient over the 2 tested"* — PMID: 38977084
>
> *"Whole-blood mRNA analysis revealed a strong interferon signature"* — PMID: 25842288

Two siblings with a *PRKDC* defect presenting with cutaneous granulomas and recurrent lung infections (PMID: 30121298) further illustrate that primary immune deficiency in this disorder may first present with skin findings, and that modifier genes and environmental factors likely shape the variable phenotype.

F003 — DNA-PKcs deficiency uniquely causes severe human neurodevelopmental disease

A patient with **compound-heterozygous *PRKDC* mutations** (low DNA-PKcs expression, barely detectable kinase activity) had SCID plus dysmorphism, severe growth failure, **microcephaly, seizures, and profound, globally impaired neurological function**. MRI demonstrated cortical and hippocampal dysplasia with **progressive atrophy over two years**. These

neurological features were markedly more severe than in deficiencies of other NHEJ proteins (e.g., Artemis, LIG4, Cernunnos), and—critically—are **not seen in Prkdc-deficient mice, dogs, or horses**, pointing to a stringent, human-specific requirement for DNA-PKcs during neuronal development.

> *"the patient was also dysmorphic; showed severe growth failure, microcephaly, and seizures; and had profound, globally impaired neurological function"* — PMID: 23722905
>
> *"These neurological features were markedly more severe than those observed in patients with deficiencies in other NHEJ proteins"* — PMID: 23722905
>
> *"our findings demonstrate a stringent requirement for DNA-PKcs during human neuronal development"* — PMID: 23722905

The neurological severity is thought to reflect the role of NHEJ in repairing endogenous double-strand breaks in post-mitotic and developing neurons; it distinguishes DNA-PKcs deficiency from the otherwise overlapping NHEJ-SCID disorders.

F004 — Causal chain: *PRKDC* loss impairs NHEJ/V(D)J recombination and AIRE-dependent tolerance

DNA-PKcs is a core NHEJ factor that **recruits and activates Artemis to open DNA hairpins during V(D)J recombination**. Autophosphorylation of the **ABCDE/Thr2609 cluster** switches DNA-PK from an end-protection to an end-processing mode, structurally widening the DNA-binding groove to admit Artemis for hairpin cleavage. Loss of function **blocks formation of coding and signal joints**, arresting T- and B-cell development (producing T-B-NK+ SCID) and conferring cellular radiosensitivity to double-strand-break damage.

A **second, parallel mechanistic branch** explains the autoimmunity: DNA-PK interacts with the transcription factor **AIRE (autoimmune regulator)** to promote central T-cell tolerance. Mutant DNA-PKcs failed to drive AIRE-dependent transcription of peripheral-tissue antigens in vitro, and this correlated in vivo with anti-calcium-sensing-receptor autoantibodies typical of AIRE deficiency.

> *"In mice DNA-PK also interacts with the transcription factor autoimmune regulator (AIRE) to promote central T-cell tolerance"* — PMID: 25842288

> *"mutated DNA-PKcs did not promote AIRE-dependent transcription of peripheral tissue antigens in vitro"* — PMID: 25842288
>
> *"Phosphorylation of four Thr residues in ABCDE leads to gross structural rearrangement of DNA-PK, widening the DNA binding groove for Artemis recruitment and hairpin cleavage"* — PMID: 34936881

F005 — DNA-PKcs deficiency occurs naturally across species with graded V(D)J severity; HSCT is curative

Spontaneous DNA-PKcs deficiency defines the classic **C.B-17 *scid* mouse** and occurs naturally in **Arabian horse foals** and **Jack Russell terrier dogs**. The severity of the V(D)J recombination block **inversely correlates with the residual DNA-PK activity** measured in normal fibroblasts from each species (horse most severe, mouse leakiest, dog intermediate). None of these animals develops the human neurological phenotype. **Allogeneic HSCT is the only established curative therapy** for radiosensitive SCID, but patients' intrinsic radio- and chemosensitivity mandates minimizing alkylating conditioning agents and ionizing radiation. Proof-of-concept gene therapy exists: **iPSC-based nuclease gene correction restored DNA-PK signaling and T-cell maturation** in a *prkdc*-deficient disease model.

> *"the severity of the V(D)J recombination deficits in these three examples of genetic DNA-PKcs deficiency inversely correlates with the relative DNA-PK enzymatic activity expressed in normal fibroblasts derived from these three species"* — PMID: 11489998
>
> *"Patients with the radiosensitive form of SCID may also have increased short- and long-term sensitivity to the alkylator-based chemotherapy regimens that are traditionally used for conditioning before allogeneic hematopoietic cell transplantation (HCT)"* — PMID: 20113890
>
> *"Genetic correction of the RS-SCID iPSC restored T-lymphocyte maturation, polyclonal V(D)J recombination of the T-cell receptor followed by successful beta-selection"* — PMID: 26000857

F006 — Diagnosis rests on T-B-NK+ immunophenotype, radiosensitivity, a P-nucleotide signature, and *PRKDC* sequencing

The recurrent L3062R mutation does not abolish DNA-PKcs kinase activity or DNA end-binding; diagnosis therefore relies on the **functional consequence**—insufficient Artemis activation, revealed by **long palindromic (P)-nucleotide stretches in immunoglobulin/TCR coding joints**—plus impaired overall end-joining and cellular radiosensitivity (colony-survival and γ-H2AX assays). The immunophenotype is **T-B-NK+ SCID**. Because kinase activity is retained, authors explicitly warn that a normal DNA-PK activity assay does **not** exclude a *PRKDC* defect, so **molecular sequencing of *PRKDC* is required for definitive diagnosis**. Newborn screening detects **low/absent T-cell receptor excision circles (TRECs)** on dried blood spots.

> *"The causative mutation did not affect the kinase activity or DNA end-binding capacity of DNA-PKcs itself; rather, the presence of long P-nucleotide stretches in the immunoglobulin coding joints indicated that it caused insufficient Artemis activation"* — PMID: 19075392
>
> *"the presence of DNA-PKcs kinase activity is not sufficient to rule out a defect in this gene during diagnosis and treatment of RS-SCID patients"* — PMID: 19075392
>
> *"Without treatment, severe combined immunodeficiency is usually fatal in the first year of life"* — PMID: 42299943

The P-nucleotide / microhomology signature is shared with Artemis deficiency (PMID: 18034425), consistent with DNA-PKcs and Artemis acting in the same hairpin-opening step.

F007 — Ultra-rarity, detected within broader SCID newborn screening

Human *PRKDC* deficiency is **extremely rare—only ~7 patients reported worldwide as of 2024**—making individual prevalence incalculable (far below 1 per million). It is one subtype within all-cause SCID, which arises from mutations in **at least 19 known genes**. A seven-year **Catalonia (Spain) TREC newborn-screening program of 420,263 newborns** diagnosed SCID in 8 infants plus 1 congenital athymia, giving an all-cause SCID incidence of **1:46,753 live births**. Inheritance is autosomal recessive; the recurrent p.L3062R allele raises the possibility of founder effects and/or consanguinity in reported families.

> *"Among 420,263 screened newborns, 105 screened positive (0.02%). SCID was diagnosed in eight infants and congenital athymia in one, corresponding to an overall incidence of 1:46,753 live births"* — PMID: 42079620
>
> *"Severe combined immunodeficiency is an inherited condition arising from mutations in at least 19 known genes"* — PMID: 42299943
>
> *"human mutations in PRKDC are extremely rare and the disease spectrum has not been described so far"* — PMID: 38977084

F008 — Prognosis is fatal untreated; allogeneic HSCT is curative, best after newborn-screening diagnosis

Untreated SCID is usually fatal within the first year of life, and **allogeneic HSCT is the only established cure**. In a 100-patient SCID HSCT cohort, overall survival was **68% (84% excluding first-month mortality)**; survival was significantly higher for cases identified by **newborn screening (93%)** versus clinical presentation, and better with bone-marrow graft source, matched related donor, and use of conditioning. For DNA-PKcs/radiosensitive SCID specifically, intrinsic radio- and chemosensitivity mandates minimizing alkylator conditioning and ionizing radiation. Supportive management includes immunoglobulin replacement, antimicrobial prophylaxis, protective isolation, irradiated/CMV-safe blood products, and avoidance of live vaccines—particularly relevant given the persistence of live rubella vaccine strain in granulomas. Experimental directions include gene therapy / gene correction (iPSC proof-of-concept).

> *"SCID cases identified by NBS demonstrated superior OS (93%) compared to cases identified by clinical presentation"* — PMID: 39900265
>
> *"The OS rate was 68%, increasing to 84% when excluding mortality in the first month after transplantation"* — PMID: 39900265
>
> *"minimizing exposure to alkylating agents and ionizing radiation is important for optimizing survival and minimizing late effects"* — PMID: 26055221

Report by Section

1. Disease Information

DNA-PKcs–deficient SCID is a monogenic, autosomal-recessive inborn error of immunity in the **NHEJ / V(D)J-recombination-defect ("radiosensitive SCID", RS-SCID)** family. The causal gene is **PRKDC**, encoding **DNA-PKcs**, the ~469 kDa catalytic subunit of the DNA-dependent protein kinase holoenzyme (DNA-PK = Ku70/Ku80 + DNA-PKcs).

Identifier	Value
MONDO	MONDO:0014423
OMIM	615966 (Immunodeficiency 26, IMD26)
Gene	*PRKDC* (HGNC:9413), chromosome 8q11.21
Protein	DNA-PKcs (UniProt P78527)
Inheritance	Autosomal recessive
MeSH	Severe Combined Immunodeficiency (D016511)
ICD-10 / ICD-11	D81.x (combined immunodeficiencies) / 4A01.1 family
Orphanet	Within "Radiosensitive SCID" grouping

Synonyms: SCID due to DNA-PKcs deficiency; PRKDC deficiency; DNA-PKcs deficiency; radiosensitive T-B-NK+ SCID due to *PRKDC* mutation; Immunodeficiency 26.

Most of the disease-level knowledge is derived from **aggregated case reports and small series** (the comprehensive review covering 7 patients, PMID: 38977084) rather than from EHR-scale cohorts, reflecting extreme rarity.

2. Etiology

Primary cause — genetic. Biallelic (homozygous or compound-heterozygous) pathogenic variants in *PRKDC*. The recurrent **p.L3062R** hypomorphic missense allele accounts for 6 of 7 reported patients (PMID: 38977084); a distinct compound-heterozygous genotype with very low expression and near-absent kinase activity underlies the severe neurological case (PMID: 23722905).

Genetic risk factors. The disease is fully penetrant Mendelian; "risk" is conferred by inheriting two pathogenic alleles. Consanguinity and possible founder effects likely contribute to recurrence of L3062R. **Modifier genes, epigenetic, and environmental factors** are proposed to explain the broad phenotypic variability (PMID: 30121298).

Environmental risk / protective factors. No classical environmental risk factors cause the disease. However, environmental exposures strongly modulate outcome: **ionizing radiation and alkylating chemotherapy are hazardous** owing to intrinsic radiosensitivity, and **live-attenuated vaccines (e.g., rubella, BCG, rotavirus) are dangerous** and can cause disseminated vaccine-strain disease and granulomas (PMID: 38977084). Protective measures are behavioral/clinical (isolation, prophylaxis, avoidance of live vaccines and radiation) rather than genetic.

Gene–environment interactions. The defective DNA-repair genotype converts otherwise routine exposures—radiologic imaging, conditioning chemotherapy, live vaccines—into serious hazards, the defining gene–environment interaction of radiosensitive SCID.

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset / severity / frequency
Severe/recurrent infections	Clinical sign	HP:0002719, HP:0002721	Neonatal–infancy; severe; near-universal
T-B-NK+ severe combined immunodeficiency	Lab abnormality	HP:0004430	Congenital; severe; defining
Granulomas (cutaneous/ visceral)	Physical manifestation	HP:0032252, HP:0001061	Infancy–childhood; variable; 5/7 (atypical form)
Autoimmunity (cytopenias, thyroiditis)	Clinical sign	HP:0002960, HP:0002633	Childhood; variable; frequent in leaky form
Interferon signature (blood)	Lab abnormality	—	Correlates with inflammation
Microcephaly	Physical manifestation	HP:0000252	Congenital; severe; subset (neurological case)
Seizures	Clinical sign	HP:0001250	Infancy; severe; subset
Progressive cerebral/ hippocampal atrophy	Imaging	HP:0002059, HP:0002543	Progressive over ~2 y; subset
Growth failure / dysmorphism	Physical	HP:0001508, HP:0001999	Congenital; severe; subset
Declining T- and B-cell counts	Lab abnormality	HP:0005403, HP:0010976	Progressive in leaky form

Quality-of-life impact: profound. Untreated disease is lethal in infancy; survivors require lifelong immune support until/after HSCT. Neurological involvement, where present, causes severe lifelong disability.

4. Genetic / Molecular Information

- **Causal gene:** *PRKDC* (HGNC:9413; OMIM 600899), 8q11.21; protein DNA-PKcs (UniProt P78527), a PIKK-family serine/threonine kinase with FAT, kinase, and FATC domains and a large HEAT-repeat N-terminal solenoid.
- **Pathogenic variants:**
- **p.L3062R** — recurrent hypomorphic missense in the FAT/kinase region; LOF for Artemis activation but retains autophosphorylation/kinase activity; classified pathogenic (functional + segregation evidence). 6/7 patients.
- **Compound-heterozygous** genotype with reduced expression + minimal kinase activity → severe SCID + neurodevelopmental disease (PMID: 23722905).
- **Variant types:** missense (predominant); other LOF classes expected but under-reported given rarity.
- **Allele frequency:** pathogenic alleles are vanishingly rare in gnomAD; the biallelic disease is <1 per million.
- **Origin:** germline, autosomal recessive.
- **Functional consequence:** loss of function with respect to Artemis activation / NHEJ end-processing (hypomorphic, not complete null, for L3062R).
- **Modifier genes / epigenetics:** proposed to shape variable expressivity (PMID: 30121298); no specific modifier locus established.
- **Chromosomal abnormalities:** none characteristic; DNA-PKcs loss predisposes to genomic instability and telomere-fusion phenotypes in model cells (PMID: 15162023, PMID: 38407308).

5. Environmental Information

- **Environmental factors:** ionizing radiation and radiomimetic/alkylating agents cause excess cellular damage owing to the DSB-repair defect (PMID: 20113890).
- **Lifestyle factors:** not applicable (congenital genetic disease).
- **Infectious agents:** opportunistic and common pathogens cause severe infection due to immunodeficiency; **live-attenuated vaccine strains** (notably **rubella virus**) can persist and drive granulomas (PMID: 38977084).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic *PRKDC* mutation (e.g., p.L3062R)** → expression of a DNA-PKcs protein that retains kinase activity but is defective in activating Artemis (*leads to*, demonstrated biochemically; PMID: 19075392).
2. During V(D)J recombination, RAG1/2 generate hairpin-sealed coding ends → DNA-PKcs normally autophosphorylates its **ABCDE/Thr2609 cluster**, rearranging to admit Artemis (*results in* hairpin opening; PMID: 34936881).
3. Mutant DNA-PKcs **fails to activate Artemis** → hairpins are not opened, coding joints form with **long P-nucleotide stretches**, and overall NHEJ end-joining is impaired (*leads to*; PMID: 19075392).
4. Failed V(D)J recombination **arrests T- and B-lymphocyte development → T-B-NK+ SCID** (*results in*, demonstrated by immunophenotype).
5. In parallel, impaired NHEJ leaves cells unable to repair radiation/chemotherapy-induced DSBs → **cellular radiosensitivity** (*results in*; PMID: 19858715).
6. **Branch — tolerance:** mutant DNA-PKcs fails to support **AIRE-dependent transcription of peripheral-tissue antigens**, impairing central T-cell tolerance → **autoimmunity, autoantibodies, granuloma** (*leads to*; PMID: 25842288); amplified by a **type I interferon signature** and, in some, **persistent rubella vaccine strain** in tissue (*contributes to* granuloma; PMID: 38977084).
7. **Branch — neurodevelopment (human-specific):** loss of DNA-PKcs-dependent repair of endogenous DSBs in developing/post-mitotic neurons → **microcephaly, seizures, progressive cerebral/hippocampal atrophy** (*results in*; demonstrated clinically but mechanistically inferred; PMID: 23722905).

```
PRKDC mutation (L3062R)
        │  (hypomorphic: kinase OK, Artemis activation lost)
        ▼
NHEJ end-processing defect ──► cellular radiosensitivity (DSB repair failure)
        │
        ├──► V(D)J recombination block ──► T-B-NK+ SCID ──► severe infections
        │
        ├──► impaired AIRE-dependent tolerance ──► autoimmunity + granuloma
        │          (+ IFN signature, + persistent rubella vaccine strain)
        │
        └──► [human-specific] neuronal DSB repair failure ──► microcephaly,
                   seizures, progressive cerebral atrophy
```

- **Molecular pathways:** NHEJ DSB repair; V(D)J recombination; DNA-PK/PIKK signaling (GO:0006303 DSB repair via NHEJ; GO:0033151 V(D)J recombination).
- **Protein dysfunction:** loss of function for Artemis activation; structurally localized to FAT/kinase region (PMID: 35801871, PMID: 36270581).
- **Cellular processes:** arrested lymphocyte development; genomic instability; defective repair response to DSBs.
- **Immune involvement:** profound combined immunodeficiency (GO:0002250 adaptive immune response) plus dysregulated inflammation/autoimmunity.
- **Cell types (CL):** T cells (CL:0000084), B cells (CL:0000236), developing thymocytes (CL:0000893), neurons (CL:0000540).

7. Anatomical Structures Affected

- **Primary:** hematolymphoid system — bone marrow (UBERON:0002371), thymus (UBERON:0002370), lymph nodes (UBERON:0000029), spleen (UBERON:0002106).
- **Secondary / inflammatory:** skin (UBERON:0002097, granulomas), lung (UBERON:0002048, infections/granulomas), thyroid (UBERON:0002046, autoimmune thyroiditis), and other viscera with granulomas.
- **Nervous system (subset):** cerebral cortex (UBERON:0000956), hippocampus (UBERON:0002421) — dysplasia and progressive atrophy.
- **Subcellular (GO CC): nucleus** (GO:0005634) — site of DNA-PK action; DNA-dependent protein kinase complex (GO:0005958).
- **Lateralization:** systemic/bilateral; CNS atrophy global.

8. Temporal Development

- **Onset:** congenital immune defect; classic SCID presents in the **neonatal period** / **first months** with infections and failure to thrive. The leaky/granulomatous form may present later in **childhood**. Neurological features (subset) are congenital and **progressive**.
- **Progression:** without HSCT, relentless and fatal in infancy. In leaky disease, **T- and B-cell counts decline over time**, and inflammatory/autoimmune disease accrues. CNS atrophy progressed over ~2 years in the reported neurological case.
- **Critical period:** the window for pre-symptomatic newborn-screening detection and early HSCT is decisive for survival.

9. Inheritance and Population

- **Inheritance:** autosomal recessive; complete penetrance for biallelic pathogenic genotypes.
- **Expressivity:** highly variable (classic SCID ↔ granulomatous/autoimmune ↔ severe neurodevelopmental), attributed to allele type and proposed modifiers.
- **Epidemiology:** ultra-rare — ~7 patients worldwide; individual prevalence <1/1,000,000. All-cause SCID incidence context: **1:46,753 live births** (Catalonia TREC screening; PMID: 42079620); SCID arises from ≥19 genes (PMID: 42299943).
- **Founder/consanguinity:** recurrence of p.L3062R suggests possible founder/consanguinity contributions.
- **Sex ratio** / **geography:** no established predilection given tiny numbers.

10. Diagnostics

- **Immunophenotype: T-B-NK+ SCID** by flow cytometry; low/absent T and B cells, preserved NK cells.
- **Newborn screening: low/absent TRECs** on dried blood spots (multiplex TREC/KREC qPCR) — the frontline population screen (PMID: 42079620, PMID: 42496450).
- **Radiosensitivity testing:** colony-survival assay (gold standard), γ-H2AX foci — confirms the DSB-repair defect (PMID: 19858715).
- **V(D)J coding-joint analysis: long palindromic (P)-nucleotide stretches** (shared signature with Artemis deficiency) indicate impaired hairpin opening (PMID: 19075392, PMID: 18034425).
- **Caveat:** DNA-PK kinase activity may be **normal** (L3062R) — a normal assay does not exclude *PRKDC* defect (PMID: 19075392).

- **Genetic testing (definitive):** *PRKDC* sequencing via SCID/IEI NGS panels or WES/WGS (PMID: 35303369).
- **Differential diagnosis:** other radiosensitive SCIDs — Artemis/*DCLRE1C* (PMID: 19953608), *LIG4* syndrome (PMID: 16585603), Cernunnos/*NHEJ1* (PMID: 37842439); also non-radiosensitive SCID (RAG1/2, IL2RG, ADA, etc.).

11. Outcome / Prognosis

- **Untreated:** usually fatal in the first year of life (PMID: 42299943).
- **With HSCT:** overall survival ~68% (84% excluding first-month mortality) in a 100-patient SCID cohort; **93% when diagnosed by newborn screening** (PMID: 39900265). Better outcomes with matched related donor, bone-marrow source, and conditioning—balanced against radiosensitivity.
- **Complications:** disseminated infection, granuloma/autoimmunity, and—where present—neurological disability that HSCT does not reverse.
- **Prognostic factors:** age at diagnosis (newborn screening vs symptomatic), donor match, conditioning intensity/toxicity, and presence of neurological disease.

12. Treatment

- **Curative — allogeneic HSCT** (NCIT: Allogeneic Hematopoietic Stem Cell Transplantation) is the standard of care, with **radiosensitivity-adapted, reduced-genotoxic conditioning** and avoidance of ionizing radiation (PMID: 26055221, PMID: 31392129, PMID: 20113890). Haploidentical approaches with modified conditioning/GVHD prophylaxis have been used in radiosensitive SCID (PMID: 37842439).
- **Supportive care:** immunoglobulin replacement, antimicrobial prophylaxis (PJP, antifungal, antiviral), protective isolation, **irradiated/CMV-safe blood products**, and **avoidance of live vaccines** (NCIT: Supportive Care; Immunoglobulin Therapy; Antimicrobial Prophylaxis).
- **Experimental: gene therapy / gene correction** — nuclease-based correction of *prkdc* in iPSCs restored DNA-PK signaling and T-cell maturation, a proof of concept (PMID: 26000857).
- **Pharmacogenomics:** the radiosensitive genotype itself dictates avoidance of alkylators/radiation—genotype-guided conditioning.

13. Prevention

- **Primary prevention:** not applicable (congenital genetic disease); **genetic counseling** and reproductive options (carrier testing, PGD/prenatal testing) for at-risk families.
- **Secondary prevention: newborn screening (TREC)** for presymptomatic detection, enabling early HSCT and markedly improved survival (PMID: 39900265, PMID: 42079620).
- **Tertiary prevention:** infection prophylaxis, avoidance of live vaccines and radiation, and management of autoimmunity/granuloma.
- **Immunization caveat: live-attenuated vaccines are contraindicated**; inactivated vaccines are generally ineffective pre-immune reconstitution.

14. Other Species / Natural Disease

Naturally occurring DNA-PKcs deficiency is well documented:

Species	NCBI Taxon	Model/ natural	Severity of V(D)J block
Mouse (*Mus musculus*, C.B-17 *scid*)	10090	Spontaneous	Leakiest
Dog (*Canis familiaris*, Jack Russell terrier)	9615	Natural	Intermediate
Horse (*Equus caballus*, Arabian foal)	9796	Natural	Most severe

Severity inversely correlates with residual fibroblast DNA-PK activity (PMID: 11489998). **None of these animals develops the human neurological phenotype**, underscoring the human-specific neuronal requirement (PMID: 23722905). Orthologous gene: *Prkdc* in mouse/dog/horse. These are of veterinary importance (fatal SCID in affected foals and pups) and serve as comparative-biology and preclinical models. No zoonotic transmission (genetic disease).

15. Model Organisms

- **Mouse:** C.B-17 *scid* (spontaneous *Prkdc* mutation); recapitulates T-B-NK+ SCID, radiosensitivity, V(D)J block, and genomic-instability/telomere phenotypes (PMID: 15162026, PMID: 27136939). Comparator NHEJ models: Artemis-deficient mice (leaky SCID, PMID: 12504013); *Lig4* R278H knock-in (LIG4-syndrome model, PMID: 20133615).
- **Large-animal natural models:** Jack Russell terrier dog; Arabian horse foal (PMID: 11489998).

- **Cellular / iPSC:** patient-derived and *prkdc*-deficient iPSCs corrected by targeted genome editing restored DNA-PK signaling and T-cell differentiation (PMID: 26000857).
- **Model limitation:** no model reproduces the **human neurodevelopmental disease**, the key gap between animal and human phenotypes.
- **Resources:** MGI, IMSR (mouse); OMIA (natural animal disease); Cellosaurus (iPSC/cell lines).

Mechanistic Model / Interpretation

DNA-PKcs sits at the intersection of two biological programs that together explain the full disease spectrum. First, as the **regulatory hub that licenses Artemis** to open DNA hairpins, DNA-PKcs is indispensable for V(D)J recombination; its loss produces the core T-B-NK+ SCID with radiosensitivity shared across NHEJ defects (Artemis, LIG4, Cernunnos). The recurrent **L3062R allele is informative precisely because it is "surgical"**: it preserves the kinase's catalytic and DNA-binding functions but severs the Artemis-activation step, producing a diagnostic P-nucleotide fingerprint and the warning that kinase assays cannot exclude the disease.

Second, DNA-PK's **partnership with AIRE** adds a tolerance-breaking layer absent from a simple "empty immune compartment" model, explaining why the human cohort is dominated by **granulomas, autoimmunity, and an interferon signature** rather than pure infection susceptibility. Persistent live rubella vaccine strain acts as an antigenic driver of granuloma within this tolerance-defective, immune-dysregulated milieu.

Third, and uniquely, DNA-PKcs is required for **human neuronal genome maintenance**, a requirement not met by compensatory repair in mouse, dog, or horse. This species difference is the single most striking feature of the human disease and the principal unmet gap in animal modeling.

Therapeutically, the logic is coherent: **HSCT corrects the hematopoietic defect**, newborn screening maximizes survival by enabling pre-infection transplant, and the intrinsic DSB-repair defect converts conditioning genotoxins into hazards—mandating reduced-intensity, radiation-sparing regimens. Gene correction offers a future route that could, in principle, also address non-hematopoietic (e.g., neuronal) requirements, though this remains preclinical.

Evidence Base

PMID	Role in this report
38977084	Comprehensive review/update: 7-patient spectrum, L3062R recurrence, granuloma/autoimmunity, rubella in granuloma, rarity
19823081	First human DNA-PKcs deficiency; hypomorphic L3062R retains expression/phosphorylation
35801871	Structure of Artemis:DNA-PKcs basal state; links L3062R to insufficient Artemis activation
23722905	Compound-het patient with severe neurodevelopmental disease; human-specific DNA-PKcs requirement
25842288	AIRE-dependent tolerance branch; interferon signature; autoimmunity mechanism
34936881	ABCDE/Thr2609 autophosphorylation switch for Artemis recruitment/hairpin cleavage
19075392	P-nucleotide diagnostic signature; retained-kinase caveat
11489998	Cross-species natural models; graded V(D)J severity vs residual activity
20113890	Radiosensitive SCID: chemo/radiation sensitivity and conditioning
26000857	iPSC genome-editing restores DNA-PK signaling and T-cell maturation
39900265	HSCT survival 68/84%; 93% with newborn screening
26055221	Rationale for minimizing alkylators/radiation in RS-SCID
42079620	All-cause SCID incidence 1:46,753 (TREC screening)
42299943	SCID fatal untreated in first year; ≥19 SCID genes
18034425	P-nucleotide/microhomology signature (Artemis defect, shared step)
30121298	Two PRKDC siblings with cutaneous granulomas; modifier hypothesis
19858715	DSB-repair defects and radiosensitivity; colony-survival assay

Supporting/contextual: 35303369 (SCID gene panel incl. PRKDC), 25987660 (V(D)J congenital defects), 36270581 (Artemis:DNA-PKcs dynamics), 38407308 (DNA-PK telomere control), 12504013 (Artemis mouse), 20133615 (LIG4 mouse), 16585603 (LIG4 human), 37842439 (haploidentical HSCT in RS-SCID), 31392129 / 28918671 (HSCT in PID and late effects).

Limitations and Knowledge Gaps

- **Extreme rarity (~7 patients)** precludes robust estimates of prevalence, penetrance of specific features, survival, and genotype–phenotype correlations; most evidence is from case reports/small series.
- **Genotype–phenotype relationships are uncertain:** why the same L3062R allele yields classic vs granulomatous disease (modifier genes, environment, epigenetics) is unresolved.
- **No animal model of the human neurological phenotype**, limiting mechanistic dissection of the CNS requirement.
- **Disease-specific HSCT outcomes** for *PRKDC* are not separately quantified; survival figures are extrapolated from all-cause/radiosensitive SCID cohorts.
- **Gene/cell therapy** remains preclinical (iPSC proof-of-concept); no clinical trials specific to *PRKDC*.
- **Allele-frequency and carrier data** for pathogenic *PRKDC* variants are sparse in population databases.

Proposed Follow-up Experiments / Actions

1. **International *PRKDC* registry** to pool cases, standardize phenotyping, and quantify genotype–phenotype correlations and HSCT outcomes.
2. **Modifier screens** (WGS + transcriptomics) in patients sharing L3062R but divergent phenotypes to identify genetic/epigenetic modifiers of granuloma/autoimmunity.
3. **Humanized or neuron-focused models** (patient iPSC-derived cortical organoids) to interrogate the human-specific neuronal DSB-repair requirement.
4. **Mechanistic validation** of the DNA-PK–AIRE tolerance axis in patient thymic/mTEC systems, with therapeutic implications for autoimmunity.

5. **Radiosensitivity-adapted conditioning trials** (reduced-intensity/radiation-sparing) specifically in NHEJ/RS-SCID to optimize HSCT safety.
6. **Advancement of gene correction** from iPSC proof-of-concept toward in vivo/ex vivo gene-editing strategies, addressing whether hematopoietic correction suffices given non-hematopoietic requirements.
7. **Systematic rubella/live-vaccine surveillance** in granulomatous combined immunodeficiency to guide vaccine-avoidance policy.
